

Lab: Agents in the Wild

Learning Goal: Apply the agent framework to real-world examples at different scales.

Part 1: Agent Analysis

Instructions: For each of the following, determine whether it qualifies as an agent under the Active Inference framework. For each, identify: (a) generative model, (b) predictions, (c) prediction errors, (d) actions to minimize error.

1. A venus flytrap
2. A GPS navigation app
3. A sleeping human

Write your response here

Part 2: The Prediction-Error Cycle in Detail

Learning Goal: Trace the predict-compare-update cycle through a detailed scenario.

Scenario: You walk into a coffee shop expecting to order your regular drink, but the barista tells you they're out of oat milk.

Trace the steps:

1. What was your prediction?
2. What was the prediction error?
3. What are your options: (a) update your beliefs, (b) act to change the situation? List specific examples of each.

4. Which option do you choose, and why?

Write your response here

Part 3: The Self-Model

Learning Goal: Explore how the self-model shapes experience.

Instructions: Try the following brief exercise and reflect:

1. Close your eyes and point to your right knee. How did you know where it was without looking? (This is proprioception — your body's self-model.)
2. Now, recall a time you felt a "phantom" phone vibration — your phone wasn't actually buzzing. What does this tell you about your self-model's predictions?
3. What would change in your daily experience if your self-model were disrupted?

Write your response here

Part 4: Agent Scale Comparison

Learning Goal: Compare agency across scales.

Agent	Generative Model (what it "expects")	Prediction Error Signal	Action to Minimize Error
E. coli bacterium			
A domestic cat			
A human student			
A social media recommendation algorithm			

Fill in the table, then answer: What is gained at each scale? What is lost?

Write your response here

Part 5: Reflection

In 150 words, discuss this question: Active Inference says agents "act to confirm their predictions." Does this mean we only ever see what we expect to see? How do we learn anything *new*?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Agent identification	Generative model criteria
2	Process analysis	Predict-compare-update cycle
3	Experiential exploration	Self-model and proprioception
4	Comparative analysis	Agency across scales
5	Critical reflection	Prediction confirmation vs. learning